

Electricity Theft Detection for Smart Homes with Knowledge-Based Synthetic Attack Data

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Abstract—Electricity thefts are conventionally manually detected by inspections, accusations, and the failure of meters. However, the recent evolution of machine learning may allow the automatic detection of electricity theft only from the patterns of meter readings. Electric consumption heavily relies on many factors, e.g., the lifestyle of the day and the weather, and thus the accuracy of detection is questioned. We propose an electricity theft detection framework for smart homes with knowledge-based synthetic attack data. This allows training of the attack classifier only from the legitimate power consumption data, i.e., without attack actions and associated labels. We identified five attack patterns as the knowledge which consisted of smart attacks and legacy attacks. We have conducted comprehensive evaluations with nine machine learning models using the Almanac of Minutely Power dataset version 2 (AMPds2) dataset fine-grained time-series data of a smart home. We found that Gradient Boosting-based algorithms achieved the best, and Random Forest performed alternatively with almost 100% accuracy for detecting and classifying legacy attacks. Some smart attacks were not detected, but those algorithms achieved good performance in detection and classification.

Index Terms—Electricity Theft Detection, Machine Learning, Smart Meter, Smart Home, Synthetic Data.

I. INTRODUCTION

Electricity is not visible like communications but has material values. Electricity theft attacks (ETAs) are very common across the globe [1], [2], especially in developing countries. It is said that more than a fifth of the total electricity production in India is lost due to electricity theft [3]. ETAs are conventionally detected by regular inspections of power meters and cables, users' accusations, or through meter device failures [4]. However, these methods are too dependent on humans and still have difficulties in finding electricity thefts because of the complexity of the power systems.

Electricity consumption of an appliance has its own signature in the time series, which is aggregated with the consumption of other appliances at the metering point. Thus, the output of a smart meter provides the aggregated patterns of electricity consumed from legitimate loads and malicious loads, which made it difficult to detect the attack only by heuristic analysis of the meter readings [5]. However, the evolution of machine learning makes it possible to explore the possibility of electricity theft detection from such aggregated consumption patterns.

This paper proposes an electricity theft detection (ETD) framework for smart homes with knowledge-based synthetic attack data. We identify five types of attack scenarios, which consisted of smart attacks and legacy attacks. We particularly focus on fine-grained time-series data for extracting the features of electricity loads with enough dimensions and for predictions of attack types.

One challenge with using machine learning classifiers in ETD is data imbalance, i.e., the numbers of normal and abnormal samples are not in the same range. Benign samples are easily available using historic data but attack/theft samples, on the other hand, rarely or do not exist for a given customer. Besides, in many cases, examples of attack classes cannot be obtained from the historic data due to zero-day attacks. However, the problem of imbalanced data and zero-day attacks was addressed by generating a synthetic attack dataset, benefiting from the fact that theft patterns are predictable [6]. This significantly improves the detection rate and enables the detection of a wide range of attack types.

Training an ETD machine learning model with synthetic attack data loads [7] would be a promising approach; because (1) to have attack consumption patterns along with legitimate patterns in real-life scenarios needs years and electricity costs; (2) attack-contained meter readings can be easily calculated simply by adding synthetic malicious consumption to legitimate consumption. This was not possible in computer-network communications, because the behavior of nodes depends on the existence of attacks. Along with those well-known reasons, we emphasize that (3) synthetic attack data can be based on a wide range of knowledge which may contribute to training a good ETD model for attack classifications.

The literature [7] assumed simple statistical models such as random, mean, shift, and reverse for synthetic attack loads. However, we propose to use our identified attack scenarios such as Baseload, Weakload, Peakhour, Midnight, and Evil-Twin attacks based on the discussions above. These attack dimensions are more realistic than the statistical ones because the identified attack scenarios have intentions.

Most of ETD literature [7]–[9] assume the use of smart meter data for protecting utility companies. However, the purpose of the smart meters is originally for power demand control and dynamic pricing, and thus the sampling interval

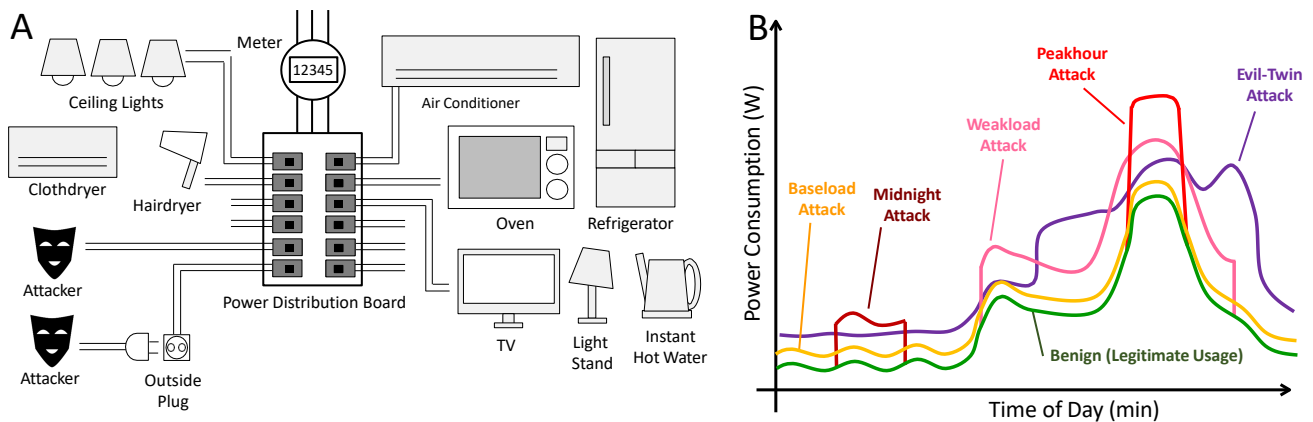


Fig. 1. A: Power Distribution Board with connected sample appliances. B: Scenarios of electricity theft attacks. Power distribution boards and cables are deployed behind walls and are usually not visible. An attacker uses the pre-deployed cable in a complex building or the outside plug for stealing electricity. Depending on the theft pattern, we identify five classes of attacks.

of them is 30 minutes or 15 minutes. Because of privacy reasons, they cannot make it shorter for reporting to the utility company. However, internally, smart meters have fine-grained data per minute or second. These data samples contain a lot of representations of the appliances installed and of households' behavior. Such fine-grained private data cannot be exported but can be used for ETA detection for the household as a smart home application.

It is worth noting that the protection of utility companies [7]–[9] and the protection of homes are different even though they both resist electricity thefts or power stealing attacks. If an attacker directly steals power from a utility company by bypassing or tampering with the meter, the home's electricity bills will be decreased. However, it is not necessarily from the grid. The attacker can steal power from another home, then the electricity bill of the home will be increased. The household needs to detect it and protect it by itself because the utility company will not care about the increase in consumption by electricity theft.

In the present study, we aim to provide a thorough comparison of the nine different ML algorithms showing that Gradient-boosting algorithms perform the best and random forest performs in alternative ways. We also study how these algorithms perform predictions by class showing that only the classification of Weakload and Peakhour attacks were difficult in these algorithms, while the other algorithms also faced difficulties in classifying the other attack types.

The rest of the paper is organized as follows. Section II discusses related works. Section III describes the attack model before proceeding to Section IV to explain our proposal: Electricity Theft Detection with synthetic attack data. Section V contains the dataset for electricity theft detection. Section VI contains the experiment results and performance evaluation. In Section VII, we provide discussions and future works, and Section VIII concludes this paper.

II. RELATED WORK

Aldegheishem et. al., [10] have identified five methods for electricity theft attacks from the viewpoint of a utility company: i.e., (1) bypassing the meter, (2) hacking the meter, (3) direct hooking, (4) tapping, and (5) meter tampering. False data injection attack (FDIA) for electricity theft [8], [9] is also discussed as a cyber-attack on the cyber-physical systems. Jokar et. al., [6] also assumes FDIA along with the physical attacks that bypass or tamper the meters, saying that all the attacks' outcome appears at the meter readings for reducing electricity consumption [11]. In this way, electricity theft detection by analyzing the meter readings was paid attention to.

Support vector machines (SVM) were studied as one of the methods for classifying electricity consumption patterns and for detecting electricity theft [6], [12], a Decision tree was also studied in [13], and Gradient-Boosting has been also studied recently by [7], [14]. The basic ideas behind these approaches are supervised learning with benign and attack data samples which ground truth labels should be given, but not easy to collect in real-life scenarios.

Anomaly detection is another approach for electricity theft detection under the scenario where labels are not available. Depending on the learning models, anomaly pattern detection based on hypothesis testing (APD-HT) [15], hierarchical self-organizing map (SOM) [16], a stacked sparse denoising autoencoder [17] has been proposed. The disadvantage of anomaly detection is to pick up any abnormal cases, such as abnormal use of appliances even if it is just turned on at the night.

Our work also assumes the real-life scenario where labels are not available. However, we introduce the knowledge of possible attack cases and synthetic attack data for training a supervised model from the non-labeled real-world dataset. Besides, we assume to secure a home from the threats of electricity theft by others. If an attacker steals electricity not from the power grid, but from other houses, the utility

company will not care about the theft, but every household needs to care about it.

The work of Punmiya et. al., [7] takes a similar approach to our work in that they propose Gradient Boosting and synthesizing the theft patterns. However, their target is to protect the utility company, not the home. Thus, the electricity theft pattern is designed to reduce the bill. Please note that in our case, theft increases the bill. The literature considers random-based reduction, mean-based reduction, and reverse-time-based reduction for training a binary classification model between benign and attack based on the past work [6]. Our paper tries to simulate more reasonable electricity theft patterns and train a multi-classification model for predicting the class of attack if detected which is novel and more challenging. Our work comes with fine-grained time-series data in a smart home environment, which is not available in the smart grid today.

III. ELECTRICITY THEFT ATTACK

A house has an electric meter for accounting purposes. The household has to pay the bill based on the accumulated power counted by the meter. An attacker may physically access the power distribution board under the electric meter and steal power from the house to make the household pay the bills (Fig. 1) on their behalf. An attacker may also steal power from an outlet if it is physically accessible outside.

This kind of scenario may happen in apartments, congested complex buildings, and/or office rooms where power line cables are not easily traceable. For example, some houses might be leased with pre-deployed lines for electricity theft.

In this paper, we have identified the following five attack classes for theft consumption patterns.

- 1) **Baseload Attack:** In the Baseload attack, the attacker continuously and constantly steals the power from the power line just as it is leaked. The instant power is not large, but accumulated theft power becomes huge. The owner of the power will not notice that the power is theft.
- 2) **Weakload Attack:** In the Weakload attack, the attacker has a sensor and steals the power weakly only when the power consumption of the home is high – in other words, when some appliances, such as TVs, washing machines, and refrigerator, are consuming the power. The owner of the house will simply misunderstand that the power consumption increases because they use their appliances.
- 3) **Peakhour Attack:** In the Peakhour attack, the attacker has a sensor and steals the power largely only when the power consumption of the home is very high – in other words, when some appliances, such as oven, heat pump, and hairdryer are consuming the power. The owner of the house will simply misunderstand like in the Weakload attack case. The attacker consumes large power during the short peak periods.
- 4) **Midnight Attack:** In the Midnight attack, the attacker steals the power at midnight when the households are sleeping. This attack can happen when a power outlet is available outside the house. They physically connect

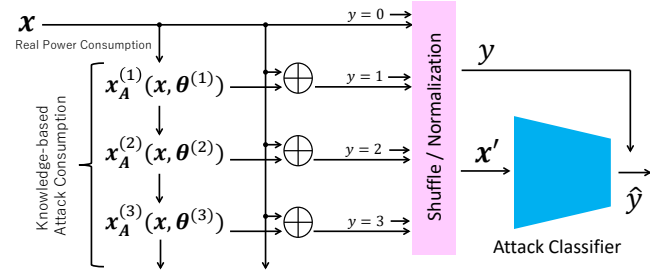


Fig. 2. The framework for electricity theft detection with synthetic attack data. Without real attack data, possible attack patterns can be theoretically generated and added onto real power consumptions for training an attack classifier.

the cable to the power outlet at midnight and steal to charge their battery including electric vehicles or to boil hot water (i.e., thermal storage).

- 5) **Evil-Twin Attack:** In the Evil-Twin attack, the attacker steals and uses the power at their house. The pattern of power consumption will look like the aggregated power consumption of two houses. In this scenario, the same kinds of home appliances will co-exist in the consumption domain. For example, two refrigerators, two washing machines, and two TVs will appear in the power consumption patterns.

IV. ELECTRICITY THEFT DETECTION WITH SYNTHETIC ATTACK DATA

To detect such electricity theft attacks, we take the approach of monitoring power consumption with enough granularity in both time and power domains at the power aggregation point. In this paper, we explore the possibility of synthetic attack data deployed to validate machine learning applications for identifying the attack types only from legitimate electricity consumption patterns.

Figure 2 shows the framework of our synthetic attack data learning for electricity theft detection.

Let us consider $\mathbf{x}$ – a vector of power consumption. This vector contains the power consumption of each timeslot in the time order. For example, $\mathbf{x}$ may represent a power consumption of a certain day, and the i -th element x_i corresponds to the power usage at i -th minute from the beginning of the day. In this case, $\mathbf{x}$ has 1440 elements: i.e., $60 \times 24 = 1440$.

In supervised learning, we assume that each $\mathbf{x}$ has a corresponding label y for training a classification model for power consumption patterns. In our case, we can assume that label $y = 0$ as a benign case, $y = 1$ as a Baseload attack, $y = 2$ as a Weakload attack, and so on.

In the real, practical scenario, we will only get a collection of $\mathbf{x}$ from a house as a result of long-term monitoring, and we will not get real attack-enabled cases with labels. However, many power-stealing cases can be simulated just by arithmetically adding stolen power as a power consumption.

Let $\mathbf{x}_A$ be a vector of stolen power by an attacker. As we assume the attacker changes the stealing power based on the

TABLE I
APPLIANCES AND UNITS IN AMPDs

Point ID	Description	Point ID	Description
WHE	Whole House Meter	FRE	HVAC/Furnace
B1E	North Bedroom	GRE	Garage
B2E	Master/South BR	HPE	Heat Pump
BME	Basement	HTE	Instant Hot Water Unit
CDE	Plugs/Lights	OFE	Home Office
CWE	Clothes Dryer	OUE	Outside Plug
DNE	Clothes Washer	TVE	Entertainment TV/PVR/AMP
	Dinning Room Plugs	UTE	Utility Room Plug
DWE	Dishwasher	WOE	Wall Oven
EBE	Electronics Workbench		
EQE	Security/Network	UNE	Unmetered Loads
FGE	Kitchen Fridge		

consumption of the house, $\mathbf{x}_A$ is a function of $\mathbf{x}$ and attacking parameters θ : e.g., $\mathbf{x}_A(\mathbf{x}, \theta)$.

Depending on the attack cases, i.e., depending on the label $y (\neq 0)$, we can consider different stolen power vectors: $\mathbf{x}_A^{(y)}(\mathbf{x}, \theta^{(y)})$.

Finally, we can get the labeled dataset as follows.

$$(\mathbf{x}', y) = \begin{cases} (\mathbf{x}, 0) & y = 0 \\ (\mathbf{x} + \mathbf{x}_A^{(y)}(\mathbf{x}, \theta^{(y)}), y) & y \neq 0 \end{cases} \quad (1)$$

Supervised machine learning models can be applied to the collection of $(\mathbf{x}', y)$.

V. DATASET FOR ELECTRICITY THEFT DETECTION

A. Overview

We have used AMPds2 (Almanac of Minutely Power Dataset version 2) [19] as a benchmark for the two-year power consumption of homes. AMPds2 consists of minute-level measured powers at the outputs of a power distribution board. They have put the names of their monitoring points as Table I.

Depending on the configuration of the houses, they may not have some appliances such as Clothes Dryers, Wall Ovens, or Dishwashers. So, by removing the power consumption of some optional appliances, we have virtually set up three types of homes, as follows.

- **Home A** is composed of full set of the appliances, which is the aggregated power consumptions of B1E, B2E, BME, CWE, DNE, HTE, EBE, EQE, FRE, OFE, OUE, TVE, UTE, CDE, HPE, DWE, FGE, and WOE. Some of those appliances such as those associated with HPE and WOE make peak power consumption, which may allow the peak-hour attacker to steal power more efficiently.
- **Home B** excludes CDE and WOE from Home A, assuming that the existence of a cloth dryer and wall oven may influence the accuracy of attacker detection.
- **Home C** excludes DWE, HPE, and TVE from Home A, assuming that the existence of a dishwasher, heat

pump and small appliances may influence the accuracy of attacker detection.

As the electric power consumption is time-dependent data, we have picked up the first 80% of data (i.e., 584 days) for the training data, applying our synthetic attack data method, and the last 20% of data (i.e., 146 days) for the test data. The test data also contains the simulated attacks in this study. Please also note that the test data is not covering the whole year, because we wanted to use a larger amount of data for training.

B. Data profiles

Table II shows the profiles of the data for our ETA detection study. We have 584 benign records for training for each home from the original monitoring data. We simulated and added Baseload, Weakload, Peakhour, Midnight, and Evil-Twin attacks based on the definitions in Section III, and the method of Section IV. For more details, Table III shows the parameters of each attack. In some conditions, if the base power consumption of the home does not reach the threshold, Weakload and Peakhour attacks are not triggered. Such attack samples are not included in the dataset.

Figure 3 shows examples of Benign and Attack data samples of different three days of Home A. From these figures, we can observe that the electric consumption pattern has a huge change by the day, and we cannot easily recognize the attack class only from a single power consumption data.

Figure 4 shows the Uniform manifold approximation and projection (UMAP) [18] of the data samples for Homes A, B, and C. Each plot corresponds to a day. We can observe that different homes have different characteristics in data. For example, in Home A, benign data is scattered having overlaps with all the types of attack samples, whereas in Home C, we can observe more clusters that might be easier for classification.

C. Attack Impact

We define attack impact (AI) as a score to measure electricity theft. This attack impact is the amount of stolen energy in the bill.

Let w_i be the weighted price of the power of x_i on the day at each index i . Let us denote the vector of w_i by $\mathbf{w}$.

The attack impact of electricity theft of the day is :

$$I = \mathbf{w} \cdot \mathbf{x}_A = \sum_i w_i x_{Ai} \quad (2)$$

Here, x_{Ai} represents the power stolen at index i of $\mathbf{x}_A$.

For calculating AI on our dataset, we have assumed 0.20 USD/kWh constantly for the unit price. The daily bill mounted by the attacker is around 1 USD on average (see, Table II). It will be about 30 USD monthly, and 360 USD in a year. If the base power consumption of the home is larger, such as in the case of an office, shop, restaurant, or factory, the attacker will be able to steal much more power.

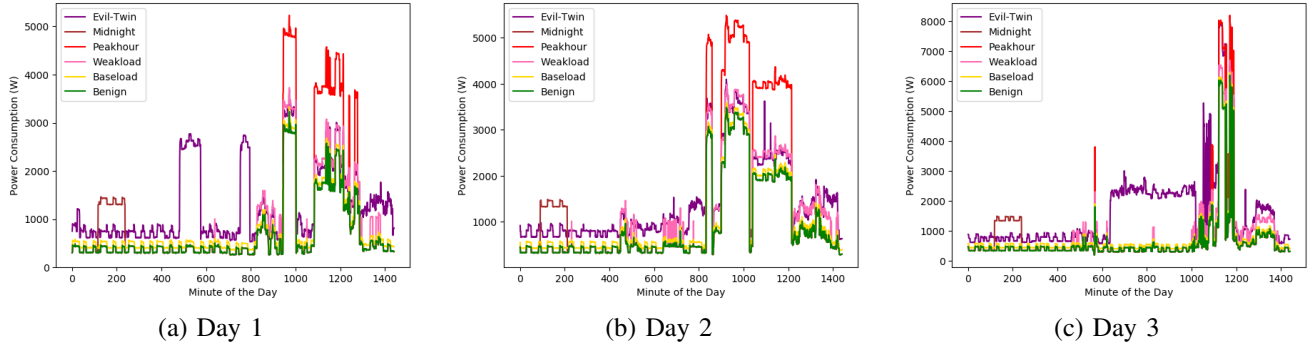


Fig. 3. Electricity usage of Home A with synthetic attack data on different days. The benign consumption patterns drastically change day by day. Especially, The Weakload and Peakhour attacks are sophisticated – having similar consumption patterns with the legitimate usage although the values are different.

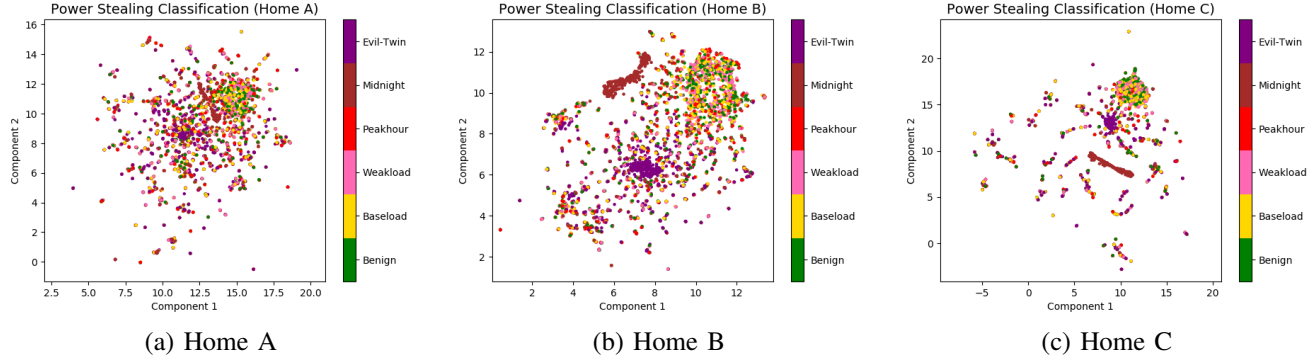


Fig. 4. 2D projections of attack contained electricity usage with the benign case by uniform manifold approximation and projection (UMAP) [18]. Each plot corresponds to the consumption pattern of a day on different attack scenarios. Depending on the available appliances of the homes, the shapes of the attack classes are very different. There are more overlaps among classes in Home A, whereas we observe fewer overlaps in Home C.

TABLE II
THE PROFILE OF THE DATASET GENERATED WITH SYNTHETIC ATTACK DATA, AND CORRESPONDING ATTACK IMPACTS (AI).

Home	Category	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin	Averaged AI	AI Ratio
A	Train	584	584	580	490	584	584	1.15 USD	1.77
	Test	146	146	146	140	146	146	1.17 USD	1.72
B	Train	584	584	580	437	584	584	1.08 USD	1.80
	Test	146	146	146	130	146	146	1.10 USD	1.77
C	Train	584	584	576	306	584	584	0.78 USD	1.66
	Test	146	146	146	84	146	146	0.76 USD	1.68

VI. EVALUATION

A. Experiment Settings

In this study, we have taken the approach of surveying a wide range of machine learning algorithms with the synthetic framework in order to find the ranking of the algorithms for our problem.

Table IV shows the configuration of the experiment in our evaluation. The details of these configurations are given below.

For Gradient Boosting (GB) classifier, we have set up (1) the learning rate which shrinks the contribution of each tree to default value = 0.1, (2) the number of boosting stages to perform $n_{estimator}$ to default value = 100.

For Extreme Gradient Boosting (XGB), we have used an open-source library 'xgboost' [20] that provides an efficient and effective implementation of the gradient boosting algo-

rithm. We used the python library to run Extreme Gradient Boosting to predict power output using the default extreme gradient boost classifier model: `objective="multi:softprob"`, `random_state=0`.

For Random Forest (RF) estimator, we used the default hyperparameter values with $n_{estimator} = 100$.

For Multilayer Perceptrons (MLPs), we have tested with 3-layer MLP (MLP-3) and 4-layer MLP (MLP-4). For MLP-3, we have set 720 features for the sole hidden layer. For MLP-4, we have set 720 features for the first hidden layer and 200 features for the second hidden layer. We have run 500 epochs for training. We used the default values of sklearn for other hyperparameters. For Decision Tree Classifier (DTC) we used `random_state=12` using the Sklearn library.

For Logistic Regression (LR) and Ridge Regression Classifier (RRC), we have used

sklearn.preprocessing.StandardScaler (*, copy=True, with_mean=True, with_std=True) default parameters to avoid the dataset behaving badly if the individual feature does not more or less look like standard normally distributed data to machine learning estimators.

For SVM, we used the default hyperparameter, error regularization, $c = 1000$, preventing overfitting with $\gamma = 0.01$ and defining the function to transform the dataset, $\text{kernel} = \text{'rbf'}$.

For carrying out these experiments, we have used Google Colaboratory.

B. Performance Overview

Table V shows the ranking of the algorithms based on the accuracy of Home A. GB and XGB performed the best in every home scenario. RF could also achieve higher accuracy than the others. Compared to them, MLPs did not perform well. We found that SVM, LR, and RRC do not fit our electricity theft dataset well.

In many algorithms except RF, GB, and XGB, we have observed the performances of homes as Home C > Home B > Home A. There were about 5% to 15% gaps between their accuracies. This is probably because Home A contains a lot of legitimate consumption peaks, whereas Homes B and C do not. This indicates that the detection of electricity theft in Home A is potentially more difficult than in Homes B and C.

However, RF, GB, and XGB could successfully achieve high accuracy in Home A although it is still lower than the other homes. These results indicate that RF, GB, and XGB, especially GB and XGB are promising algorithms for detecting electricity theft with smart meters.

C. Performance by Attack Class

Figure 5 shows the confusion matrices of the trained models for Home A. The colors of this figure are:

- Red if the overall accuracy is more than 80%.
- Green if the overall accuracy is between 70% and 80%.
- Blue if the overall accuracy is between 60% and 70%.
- Gray if the overall accuracy is less than 60%.

The numbers (i.e., No. #) indicate the rank, which is corresponded to Table V.

We can observe that Midnight and Evil-Twin attacks were well classified in all the classifiers. Especially, with GB,

TABLE IV
THE CONFIGURATION OF THE EXPERIMENT.

Model	Library	Configuration
GB	sklearn	learning rate = 0.1 n_estimator=100
XGB	xgboost	objective="multi:softprob" random_state=0
RF	sklearn	n_estimator=100
DTC	sklearn	random_state=12
MLP-4	sklearn	hidden_layer_1=720 hidden_layer_2=200 epoch=500
MLP-3	sklearn	hidden_layer_1=720 epoch=500
LR	sklearn	copy=True with_mean=True with_std=True
RRC	sklearn	copy=True with_mean=True with_std=True
SVM	sklearn	c=1000 gamma=0.01 kernel='rbf'

TABLE V
EXPERIMENTAL RESULTS OF DIFFERENT ALGORITHMS. THE RANKING IS BASED ON THE ACCURACY OF HOME A.

Ranking	Models	Home	Accuracy	Precision	Recall	F1-score
1	Gradient Boosting(GB)	A	92.98%	0.94	0.93	0.93
		B	94.53%	0.95	0.95	0.95
		C	93.61%	0.94	0.94	0.93
2	Extreme Gradient Boosting(XGB)	A	92.53%	0.94	0.93	0.92
		B	95.00%	0.96	0.95	0.95
		C	92.75%	0.94	0.93	0.92
3	Random Forest (RF)	A	86.78%	0.91	0.87	0.87
		B	89.19%	0.92	0.89	0.90
		C	88.94%	0.93	0.89	0.88
4	Decision Tree Classifier(DTC)	A	70.58%	0.75	0.71	0.71
		B	77.33%	0.81	0.77	0.78
		C	80.47%	0.82	0.80	0.80
5	Multilayer Perceptron 4 (MLP-4)	A	66.44%	0.70	0.66	0.67
		B	71.28%	0.76	0.71	0.72
		C	77.52%	0.78	0.78	0.78
6	Multilayer Perceptron 3 (MLP-3)	A	65.75%	0.69	0.66	0.66
		B	69.88%	0.73	0.70	0.71
		C	71.01%	0.76	0.71	0.72
7	Logistic Regression (LR)	A	54.83%	0.54	0.55	0.54
		B	56.86%	0.56	0.57	0.55
		C	70.76%	0.73	0.71	0.71
8	Ridge Regression Classifier(RRC)	A	52.30%	0.53	0.52	0.52
		B	54.88%	0.56	0.55	0.54
		C	62.04%	0.67	0.62	0.62
9	Support Vector Machine(SVM)	A	52.29%	0.51	0.52	0.51
		B	56.16%	0.56	0.56	0.55
		C	67.69%	0.69	0.68	0.67

TABLE III
THE CONFIGURATION OF SYNTHETIC ATTACK DATA FOR SUPERVISED LEARNING IN THE EXPERIMENT.

Attack Class	Configuration
Baseload	Theft = 100W
Weakload	Threshold=500W Theft=500W
Peakhour	Threshold=1500W Theft=2000W
Midnight	Starting from 1 AM - 2 AM (at random) Duration = 120 minutes Theft=1000W
Evil-Twin	One day randomly chosen in the dataset.

XGB, and RF – i.e., the best three algorithms, have achieved 100% accuracy for these attacks. Compared to these attacks, Baseload, Weakload, and Peakhour attacks were difficult to classify. For example, we can observe confusion in classifying these attacks with Benign samples in DTC, MLP, LR, RRC, and SVM. Especially, the false positives of these algorithms were not good. For example, in LR, 33% of the Benign cases were predicted as Weakload attacks. In MLPs, 34% of the

Benign	0.95	0.00	0.03	0.02	0.00	0.00
Baseload	0.01	0.99	0.00	0.00	0.00	0.00
Weakload	0.08	0.00	0.92	0.00	0.00	0.00
Peakhour	0.26	0.00	0.01	0.72	0.00	0.00
Midnight	0.00	0.00	0.00	0.00	1.00	0.00
Evil-Twin	0.00	0.00	0.00	0.00	0.00	1.00
Actual Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin
Predicted Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin

(No.1) Gradient Boosting (GB)

Benign	0.97	0.00	0.01	0.02	0.00	0.00
Baseload	0.01	0.99	0.00	0.00	0.00	0.00
Weakload	0.07	0.00	0.93	0.00	0.00	0.00
Peakhour	0.34	0.00	0.01	0.65	0.00	0.00
Midnight	0.00	0.00	0.00	0.00	1.00	0.00
Evil-Twin	0.00	0.00	0.00	0.00	0.00	1.00
Actual Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin
Predicted Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin

(No.2) XG Boosting (XGB)

Benign	0.95	0.00	0.04	0.01	0.00	0.00
Baseload	0.05	0.95	0.00	0.00	0.00	0.00
Weakload	0.25	0.00	0.74	0.01	0.00	0.00
Peakhour	0.42	0.00	0.01	0.57	0.00	0.00
Midnight	0.00	0.00	0.00	0.00	1.00	0.00
Evil-Twin	0.00	0.00	0.00	0.00	0.00	1.00
Actual Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin
Predicted Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin

(No.3) Random Forest (RF)

Benign	0.68	0.05	0.21	0.05	0.00	0.00
Baseload	0.20	0.63	0.13	0.03	0.00	0.01
Weakload	0.27	0.03	0.65	0.04	0.00	0.00
Peakhour	0.35	0.05	0.14	0.46	0.00	0.00
Midnight	0.00	0.00	0.00	0.00	1.00	0.00
Evil-Twin	0.00	0.06	0.00	0.00	0.00	0.94
Actual Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin
Predicted Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin

(No.4) Decision Tree (DTC)

Benign	0.48	0.02	0.34	0.16	0.01	0.00
Baseload	0.22	0.45	0.26	0.03	0.01	0.04
Weakload	0.21	0.03	0.59	0.14	0.01	0.02
Peakhour	0.17	0.00	0.23	0.59	0.01	0.00
Midnight	0.01	0.01	0.00	0.00	0.96	0.01
Evil-Twin	0.00	0.02	0.06	0.00	0.00	0.92
Actual Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin
Predicted Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin

(No.5) Multilayer Perceptron 4 (MLP-4)

Benign	0.47	0.06	0.34	0.12	0.01	0.01
Baseload	0.14	0.49	0.27	0.03	0.01	0.06
Weakload	0.19	0.07	0.68	0.03	0.01	0.01
Peakhour	0.20	0.01	0.27	0.49	0.01	0.01
Midnight	0.00	0.08	0.00	0.01	0.90	0.01
Evil-Twin	0.00	0.03	0.07	0.00	0.00	0.90
Actual Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin
Predicted Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin

(No.6) Multilayer Perceptron 3 (MLP-3)

Benign	0.42	0.10	0.33	0.13	0.00	0.01
Baseload	0.14	0.34	0.34	0.11	0.00	0.07
Weakload	0.18	0.08	0.53	0.13	0.00	0.08
Peakhour	0.22	0.09	0.37	0.16	0.03	0.14
Midnight	0.01	0.01	0.01	0.00	0.95	0.03
Evil-Twin	0.00	0.06	0.05	0.01	0.00	0.88
Actual Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin
Predicted Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin

(No.7) Logistic Regression

Benign	0.57	0.16	0.17	0.08	0.00	0.02
Baseload	0.45	0.24	0.15	0.09	0.00	0.07
Weakload	0.28	0.12	0.43	0.12	0.00	0.05
Peakhour	0.28	0.12	0.26	0.22	0.00	0.12
Midnight	0.01	0.07	0.05	0.03	0.84	0.01
Evil-Twin	0.00	0.05	0.07	0.05	0.00	0.83
Actual Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin
Predicted Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin

(No.8) Ridge Regression (RRC)

Benign	0.40	0.17	0.26	0.14	0.00	0.02
Baseload	0.18	0.37	0.23	0.12	0.00	0.10
Weakload	0.21	0.14	0.43	0.14	0.00	0.08
Peakhour	0.21	0.14	0.24	0.18	0.01	0.23
Midnight	0.00	0.03	0.01	0.01	0.88	0.06
Evil-Twin	0.01	0.08	0.04	0.02	0.00	0.86
Actual Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin
Predicted Values	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin

(No.9) Support Vector Machine (SVM)

Fig. 5. Confusion matrices of the trained models. The number (No. #) indicates the rank of the overall accuracy (table V). The colors indicate the group of accuracy. Midnight and Evil-Twin attacks were relatively easily detected by all the models, especially achieving 100% accuracy with GB, XGB, Baseload, Weakload, and Peakhour attacks were difficult to classify but GB and XGB could perform well. The false positive rates of GB, XGB, and RF were about 5% which is much better than other models.

Benign cases were predicted as Peakhour attacks. These false positives should be critical in real operational scenarios. GB and XGB also have false positives in Weakload and Peakhour attacks. However, the ratio has decreased to less than 3%. They have false negatives in Peakhour and Weakload attacks with 26-34% and 7-8% respectively, which can be eventually detected if they continuously perform the attacks every day onwards.

VII. DISCUSSION AND FUTURE WORKS

In this paper, we have proposed ML model training with synthetic attack data based on the knowledge for detecting and classifying electricity theft attacks for smart homes.

On this framework, we found that Gradient Boosting gives promising results over our comprehensive evaluation of various machine learning models.

In future work, we can define other types of attacks along with the five attacks we defined. We can also study other homes or include other electricity consumption patterns for Evil-Twin attacks. The accuracy of detection can be also influenced by attack impact. If the attack impact is large, it should be detected more, but if the impact is low, it cannot be detected, but the actual impact might be negligible. This needs to be studied more in the future.

We might need to consider some data augmentation methods, especially in the case of short monitoring duration. We will also need to consider the seasonal effect such as the use of heat pumps. The use of electricity is somehow unique to the home depending on family structure, size of the homes, geographical locations, and hobbies. As the dataset is full of privacy, one of the clues for this issue can be a personalization of the general model trained by federated learning [21].

VIII. CONCLUSION

We proposed electricity theft detection for smart homes with knowledge-based synthetic attack data. For this, we have defined five types of attacks that steal electricity from other households. According to the evaluation of the AM-Pds2 dataset - minute-level two-year monitoring data of a home, Gradient Boosting-based algorithms achieved the best accuracy at 93%, and Random Forest achieved alternatively 87%. With these algorithms, Baseload, Midnight, and Evil-Twin attacks, which we categorize as legacy attacks, were successfully detected with almost 100% accuracy. However, Weakload and Peakhour attacks, which categorize as smart attacks, were not fully detected. Although we still have to study some more cases, our evaluation results indicate that our approach is useful for detecting and classifying electricity theft attacks in practical smart home environments.

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